

KERALA VETERINARY AND ANIMAL SCIENCES UNIVERSITY



RESEARCH PROJECT
ON
**FORMULATION AND STUDY OF A FROZEN DESSERT FROM
HEMP MILK**
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DECLARATION

We hereby declare that this research project entitled “FORMULATION AND STUDY OF A FROZEN DESSERT FROM HEMP MILK” is a bonafide record of research done by us during research and that research has not previously formed for the award of any degree, diploma, fellowship or other similar title, of any other university.

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We, the undersigned members of the advisory committee of research project (Group 4) agree that this research project “Formulation and Study of a Frozen Dessert from Hemp Milk” may be submitted by Group 4 (Shahanas P, Shehnas S, Sonal, Susan Eldhose, Wafa Navas) in partial fulfilment of the requirement for the degree.

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1) INTRODUCTION

The increasing awareness of sustainable food production and the demand for nutrient-dense plant-based alternatives have sparked significant interest in non-dairy products. Among these, hemp milk—a nutrient-rich, plant-based milk alternative derived from the seeds of the hemp plant (*Cannabis sativa*)—has garnered attention for its unique health benefits and environmental sustainability. Hemp milk is rich in essential fatty acids, proteins, vitamins, and minerals, making it a promising base for various food applications.

Frozen desserts, a category that includes ice cream, sorbets, and other chilled delicacies, are enjoyed worldwide and represent a versatile platform for introducing innovative ingredients. Traditional frozen desserts are predominantly made from dairy, which poses challenges for consumers with lactose intolerance, dairy allergies, or those following vegan diets. The use of hemp milk in frozen dessert formulations addresses these concerns while offering a sustainable alternative to conventional dairy-based options.

This thesis focuses on the preparation of frozen desserts using hemp milk as the primary ingredient. The study aims to explore the physicochemical properties, sensory attributes, and nutritional profile of these products to establish their feasibility and consumer acceptance. Additionally, the research investigates the functional properties of hemp milk, including its emulsification, freezing behaviour, and textural qualities, which are critical for the development of high-quality frozen desserts.

The broader significance of this study lies in its contribution to the growing field of sustainable and health-conscious food products. By leveraging the nutritional potential of hemp milk, this research seeks to expand the scope of plant-based foods while providing consumers with innovative options that align with modern dietary trends.

2) REVIEW OF LITERATURE

2.1 AN OVERVIEW INTO DIFFERENT VEGAN MILKS

2.1.1 Soy Milk:

Soy milk is the classic vegan milk that's been around for a really while. Of all the plant milks, soy milk may very closely resemble cow's milk in terms of nutritional content. It is low in calories and it's an excellent source of both protein and calcium. It is used in cooking and baking. It can be used to replace cow's milk in most recipes and is comparatively stable at high temperatures, thus making soya milk an ideal cooking ingredient while making sauces or savory dishes. It is also used in teas and coffees and can also be foamed. Soybeans are an excellent source of complete protein and thus a great alternative to avoid dairy beverage. Approximately an 8ounce serving of unsweetened soy milk can provides 105 calories, 6 grams of protein and 4 grams of fat. It is an incredible food that naturally contains numerous nutrients, also good source of vitamins specially vitamin B12, D and minerals such as calcium and phosphorus. Most widely available easily and a budget-friendly milk alternative.

Pros: Soy is a plant-based complete protein i.e. it contains all 10 essential amino acids as well as it is inexpensive as compared to other plant-based milks.

Cons: The flavour of soy milk is an initial hurdle, however choosing a flavoured variety may help overcome this issue. Soy milk does not naturally contain as much calcium as dairy milk that's the reason for calcium-fortification.

2.1.2 Almond Milk:

Almond milk is the most popular plant-based milk. However, this vegan milk is comparatively low in protein, contains lots of vitamins and other nutrients such as Vitamin E. Almond milk is rich in various antioxidants which could supposed to protect us against

the numerous ailments viz. cancer and heart disease. Home-made almond milk is a good source of calcium. Almond milk is a good beverage that you can drink as such or it can be drink as smoothies, shakes, teas, and coffees. It can be added into the cereal and oatmeal, cooking sweet and savory dishes. However, almond milk is a little too sweet for savory dishes and generally used for making desserts and smoothies by the people. Almond milk has a nutty taste and contains magnesium, selenium and vitamin E. An 8-ounce glass of almond milk generally contains 40 calories, 3.58 grams of fat and 1.51 grams of protein. Almond milk is easily available everywhere comparable to other vegan milk, which makes almond milk one of the best plant-based milk.

Pros: Almond milk contains omega-3 fatty acids that are important for heart health and the antioxidant vitamin E. It's naturally low in sugar and available in a variety of flavours.

Cons: Almond's milk is a nutritious whole food with protein and fat, however it is not a good source of protein and also low in calcium.

2.1.3 Cashew Milk:

Cashew milk is a new recent addition to plant milks. It is filled with heart-healthy unsaturated fats, which may be beneficial for those who are diabetic and health conscious consumers. This vegan milk has creamy consistency that can be used for making teas, coffees, cooking and baking. Now, it is available in most grocery stores. Cashew milk has a nuttier sweet taste.

Pros: Cashew milk completes 20% of our daily requirements of magnesium, it is rich source of mono- and polyunsaturated fatty acids, and it also contains potassium. This line-up of nutrients is important for heart health.

Cons: Homemade cashew milk are usually higher in fiber, protein and fat compared to the store-bought cashew milk. Cashew milk might be not a good dairy replacement due to low protein content.

2.1.4 Oat Milk:

Oat milk has a mild, slightly sweet flavour and high fiber compared to other milks. Fiber is essential in a balanced diet for healthy digestion and to decrease cholesterol levels. It is good for savoury and sweet recipes prepared from the cereals, smoothies, cream soups, and curries. Oat milk is high in soluble fibre and beta-glucans, however, it is lower in protein, vitamins and minerals compared to cow's milk. Generally, an 8-ounce glass of oat milk contains 130 calories, 2 grams of fat and 4 grams of protein³. Its creamy consistency is helpful to substitute the dairy milk in baked goods, soups, lattes and more. This plant-based milk has more carbohydrates than other options, which makes for a good pre- or post-workout energy drink.

Pros: Oat milk contains naturally occurring fiber and protein, and it's can be fortified with vitamins A, B2, B12 and D, and minerals calcium and phosphorus. It doesn't contain lactose, nuts or soy therefore, it's an allergen-friendly option. It is a safe option for those with Celiac disease, if made with certified gluten-free oats.

Cons: Some people love the thick texture and flavour of oat milk, while others find it a turn-off. Oats are a nutritious food, but drinking oat milk could not be equated with eating whole oats.

2.1.5 Rice Milk:

Rice milk is made by combining boiled rice with water and adding some sweeteners. Its sweet taste makes it perfect for deserts and its thinner consistency is suitable in soups and light weight sauces. Rice milk can also be used in baking however; due to its delicate texture it may require some thickening agents such as flour, corn-starch or xanthan gum. Rice milk is an allergy-friendly choice for those who ought to avoid nuts, seeds, soy or lactose. It's easily available in various grocery stores. Rice milk has the largest carbohydrate level of any plant-based milk on this list (excluding sweetened and flavoured

varieties), but this isn't always a bad thing unless you're watching your carb consumption for health reasons.

Pros: Rice milk is inexpensive and can be fortified with vitamins and minerals such as vitamins B12 and D, and calcium. It has a mild flavour that's lightly sweet and adaptable.

Cons: Rice milk is recognised for having a watery texture. It may contain added sugars, gums and additives for flavour, consistency and shelf-life. It is not a protein-rich food.

2.1.6 Coconut Milk:

Coconut milk is available in two different forms, either in a can or as a beverage. Those utilized in cooking typically come in a can. It adds a delicious coconutty flavour to any dish in which it is used and works great when used in baking too. Coconut milk is ideal for different types of recipes such as curries, soups, stews, sauces, smoothies, puddings, and even ice creams. The coconut milk that is used for drinking has a thinner consistency and even can be used for teas, coffees, and cereals. Coconut milk is less of a drinking beverage and more of an ingredient for cooking and is easily available.

Pros: Coconut milk is rich in minerals, such as manganese, magnesium, potassium, copper and selenium and contains medium-chain triglycerides. It is high in fiber compared to other plant-based milks. Wide varieties of coconut milk are available for different culinary needs, such as drinks, sauces, baking, cooking.

Cons: It is high in calories and fat, low in protein. Again, this is not inherently unhealthy, but it could be a con for some, depending on personal health needs, goals and lifestyle choices.

(Sunidhi et al., 2021)

2.2 COMPARISON OF VARIOUS TYPES OF VEGAN MILK WITH COW'S MILK

Table 2. Comparison of nutritive potential of bovine milk with commercially available nondairy plant-based milk substitutes.

Type of milk	Carbohydrate (g/100g)	Protein (g/100g)	Fat (g/100g)	Fiber* (g/100g)	Energy (K Cal)	Minerals (mg/100g)				
						Fe	Ca	K	Na	P
Bovine milk*	3.20–5.40	2.90–6.00	3.40–6.40	Not reported	69.0–118.0	0.07–0.08	122.0–134.0	152.0–181.0	41.0–58.0	119.0–121.0
Soy milk	4.64–4.92 (144.99–91.11)	3.82–3.98 (131.72–66.33)	3.1–4.3 (91.17–67.19)	0.64–0.74	51.5–52.5 (74.64–44.49)	0.50–0.66 (714.28–825.0)	4.0–5.4 (3.28–4.03)	141.0–215.0 (107.80–118.78)	2.20–12.0 (5.36–20.69)	49.0–62.60 (41.17–51.73)
Almond milk	4.30–4.70 (134.37–87.04)	2.50–1.90 (86.21–31.66)	3.20–3.60 (94.11–56.25)	1.15–1.35	55.00–55.90 (79.71–47.37)	1.35–1.45 (1928.57–1812.5)	13.05–13.15 (10.69–9.81)	65.03–65.00 (46.1–35.91)	6.38–6.60 (15.65–11.38)	75.03–75.33 (63.05–62.25)
Oat milk (fresh)	27.30–50.01 (853.12–926.11)	9.70–17.30 (334.48–288.33)	5.20–12.40 (152.94–193.75)	11.53–20.07	576.56–607.10 (835.59–514.49)	6.40–7.40 (9142.85–9250.0)	84.30–85.60 (69.09–63.88)	669.20–671.60 (474.61–371.05)	3.11–3.21 (7.58–5.53)	672.30–816.32 (564.95–674.64)
Rice milk	9.41–12.70 (294.06–235.18)	0.28–1.26 (9.65–21.0)	0.97–1.11 (28.53–17.34)	0.30–0.65	47.00–112.0 (68.12–118.0)	0.20–0.96 (285.71–1200.0)	118.0–121.35 (96.72–90.56)	26.34–27.00 (18.68–14.92)	39.20–40.59 (95.61–69.98)	55.91–56.86 (46.98–46.99)
Cocoa milk drink (fresh)	44.00–55.00 (1375.0–1018.52)	12.00–19.00 (413.79–316.66)	20.00–25.00 (588.23–390.62)	25.00–35.00	225.0–325.0 (326.09–275.42)	10.00–15.00 (14285.7–18570.0)	100.0–180.0 (81.97–134.32)	1500.0–2000.0 (1063.82–1104.97)	Not reported	750.0–1000.0 (630.25–826.45)
Kidney bean milk	53.20–59.65 (1662.5–1104.63)	17.95–26.80 (618.96–446.66)	0.66–2.49 (19.41–38.91)	6.40–7.20	127.10–132.0 (184.20–111.86)	1.50–1.80 (2142.8–2250.0)	31.30–36.20 (25.65–27.01)	344.0–356.2 (243.97–196.79)	11.0–11.8 (26.83–20.34)	68.10–72.01 (57.23–59.51)
Peanut milk (fresh)	21.51–16.10 (672.19–298.15)	25.80–26.37 (889.65–439.50)	47.22–49.20 (1388.82–768.75)	8.00–8.50	462.78–567.0 (670.69–480.51)	2.26–4.58 (3228.57–5725.0)	54.00–92.00 (44.26–68.66)	501.00–658.0 (355.32–363.53)	6.00–64.23 (14.63–110.74)	76.00–358.0 (63.86–295.87)
Coconut milk	3.75–9.41 (117.19–174.26)	0.59–2.00 (20.34–33.33)	4.12–6.00 (121.17–93.75)	5.30–5.98	50.00–92.00 (72.46–77.96)	3.90–4.01 (5571.43–5012.5)	176.0–178.1 (144.26–132.9)	631.0–639.02 (447.52–353.05)	36.00–38.36 (87.80–66.14)	240.0–256.35 (201.68–211.86)
Hemp milk	2.50–20.00 (78.12–370.37)	0.83–4.00 (28.62–66.66)	1.25–3.00 (36.76–46.87)	Not reported	19.00–21.06 (27.54–17.84)	4.89–6.58 (6985.71–8225.0)	12.03–20.25 (9.86–15.11)	110.00–126.58 (78.01–69.93)	85.69–140.01 (209.0–241.39)	54.84–63.98 (46.08–52.88)

*Figures presented in the parenthesis (in the same row) are standard amino acid values considered as 100% for bovine milk.

**Figures presented in the parenthesis are percentage values of amino acid in comparison to bovine milk the values for which are considered as 100%.

(Sethi et al., 2016) (Sunidhi et al., 2021)

2.3 HEMP MILK

Hemp seeds are considered to be as one of the richest sources of poly-unsaturated-fatty acids such as a-linolenic acid and linoleic acid and are advisable in many health conditions while, various products like oil, meal, flour, protein powder derived from hemp seeds are commercially available in the market. Along with all essential amino acids and fatty acids, it contains lipids (25–35%), proteins (20–25%), carbohydrates (20–30%), insoluble fibers (10–15%), vitamin E (90 mg/100 gm) and minerals (phosphorus, potassium, sodium, magnesium, sulfur, calcium, iron, and zinc). Its low allergenicity and high nutritional value makes it a good alternative over dairy and also soy and other nuts milk. Cannabidiolic acid (CBDA) is the most predominant bioactive constituent present. It provides health benefits against chronic diseases such as obesity, inflammatory bowel disease, rheumatoid arthritis,

and Alzheimer's disease. Among various antinutritional factors, phytic acid (inositol hexaphosphate, IP6) and trypsin inhibitors have a major contribution.

(Andre et al., 2016)

The major protein in hemp kernel is edestin, accounting for around 70% of hemp protein. Edestin is less soluble in water or buffer with neutral or acidic pH, but soluble in a basic buffer. Despite its less solubility, edestin is known for its high digestibility. The second major hemp protein is albumin, higher protein solubility and foaming capacity. Another noteworthy protein is rich in Met and Cys, and 20 mole % of the total amino acids contain sulfur.

Hemp milk is an oil-in-water emulsion and hence is highly unstable and tend to flocculate, coalesce, and generate cream which is a challenge in industry as it leads to losses in the quality and shelf-life. Emulsifiers or stabilizers are often used to counter this situation however, usage of these chemical substances is not economical as it increases the cost of production and also some health concerns such as chronic inflammatory diseases, obesity-associated diseases, and metabolic disorders do not permit the use of these chemicals. To overcome this issue, high-pressure homogenization method is performed in which protein is adsorbed on the lipid droplet surface, and can thus reduce rancidity. Homogenization pressure determines the mechanical energy provided to disrupt the oil droplets and create new interfaces. Ultrasonic treatment can be used for the optimization of the polyphenols, where the technique helps industries economically and environmentally to reduce the usage of chemicals and extraction time and enhance shelf-life.

2.4 COMPARISON OF HEMP SEED MILK WITH OTHER PLANT MILK TYPES

Plant-based milks are naturally lactose-free, cholesterol free and animal protein-free, making them a good alternative for people fed with some special diet. In addition, thanks to the bioactive components, antioxidants, monounsaturated and polyunsaturated fats components they contain, they are considered functionally valuable products as well as nutritional value. Contrary to all these positive features, they reduce the nutritional value of foods because of antinutritional factors included. They are enhanced with vitamins and minerals because of low vitamin and mineral content.

Table 2. General composition of plant-based milks

	Hemp seed milk	Oat milk	Soy milk	Almond milk	Quinoa milk	Cow milk
Protein, %	0.83-6.96	0.78- 17.30	2.6-3.98	0.42-2.50	0.49-2.9	2.9-3.32
Fat, %	1.25-18.02	0.28-12.40	1.47-13.92	1.04-8.26	0.91-2.40	3.40-6.40
Total carbohydrate %	0.3-30.21	4.44-50.01	4.14-5.06	0.58-9.38	3.5-12	3.20-5.40
Ash, %	0.47	0.48	0.71-0.87	3.04	0.11-0.22	0.67-1.9
Energy (kcal)	19-53.09	66-607.10	43-95	15-55.90	29-62	63-118
Total sugar, %	2.5	1.27	3.65	0	9.7	3.38
Total Fiber, %	0-0.4	0.8-20.07	0.2-0.74	0-1.35	0.5	0

Table 3. Comparative overview of the essential amino acid profile of plant-based milk

		Hemp seed milk	Oat milk	Soy milk	Almond milk	Cow milk
Essential amino acid (mg/100g)*	Isoleucine	3.90-4.40	4.15-4.41	1.41-2.47	29.8-39	25-62
	Leucine	6.50-7.10	7.89-9.17	2.94-4-24	83-83.20	90-108
	Lysine	4-4.60	3.79-3.91	0.88-3.92	36.20-57.4	49-96
	Methionine	2.10-2.80	1.73-1.93	0.31-0.85	27.10-27.95	17-27
	Phenylalanine	4.6-4.90	5.46-5.48	1.86-2.79	50.90-50.55	38-56
	Threonine	2.60-3.90	3.25-4.30	0.87-2.09	30.2-43.4	23-41
	Tryptophan	0.70-0.90	3.61-4.09	0.3-0.8	13.9-13.98	N.R.
	Valine	5-5.70	5.34-6.01	1.32-2.59	38.30-73.60	33-53
In vitro based values (%)**	PDCAAS, %	63-66	58-69	86-100	23	100
	DIAAS, %	N.R.	56-67	84-90.6	N.R.	118-120
In vitro protein digestibility (%) ***	Pepsin 1h digestibility	4.70	5.55	5.96	4.82	5.06
	Pancreatin 1h digestibility	17.03	19.39	14.61	20.28	21.37
	Total 2 digestibility	21.73	24.95	20.56	25.19	26.43

Data from *Paul et al. [2], **Chalupa-Krebsdaz et al. [34], **Abelilla et al. [35], ***Martinez-Padilla et al. [31]

Table 4. Fatty acid profiles of commercial plant-based milk alternatives (g/100 g total fatty acids by product)

Compounds	Hemp seed milk	Oat milk	Soy milk	Almond milk	Quinoa milk	Cow milk
SFA	12.32	18.88	14.34	22.64	10.1	68.4
MUFA	13.94	37.26	22.64	59.14	65.48	23.8
PUFA	73.73	43.85	63.02	18.21	24.42	2.21
w-6/w-3	≈ 3.43	≈ 24.64	≈ 7.52	≈ 133	≈ 19.34	≈ 3.43

Data from Martinez-Padilla et al. [31]. SFA: saturated fatty acids; MUFA: monounsaturated fatty acids; PUFA: polyunsaturated fatty acids

Table 6. Comparison of physicochemical properties for plant milk types

Properties	Hemp seed milk	Oat milk	Soy milk	Almond milk	Quinoa milk	Cow milk
Viscosity, mPas (10s ⁻¹)	25	6.77	4.91	13.48	13.20-23.8	3.15
Whiteness Index	68.49	60.21	72.17	67.36	73.52	81.89-89.18
pH	6.81-6.92	7.16	6.8-7.39	5.72-6.92	5.9-6.5	6.42-6.83

Data from McClements et al. [29], Gram [39], Kanberoglu [30], Huang et al. [40], Jeske et al. [41], Makinen et al. [33], Lai et al. [42]

(Besir et al., 2022)

Compared to cow milk, soy milk and hemp milk are most comparable to cow's milk in terms of protein ratio. Oat milk has the highest average protein content among plant-based milk. The nutritional importance of proteins is measured by their quality, which depends on the amino acid composition, digestibility, and bioavailability. PDCAAS (protein digestibility-corrected amino acid score) and DIAAS (digestible indispensable amino acid score) introduced by WHO in 1998 and by FAO in 2013, respectively, are used to assess protein quality based on in vivo experiments. While the PDCAAS and DIAAS value closest to cow protein is in soy protein, the DIAAS value of de-hulled hemp seed protein has been reported as 66.5%. Materials with a DIAAS value of 75 are qualified as a "good protein source," while those with a value above 100 are considered an "excellent source of protein" by the FAO. While cow's milk has >100% DIAAS values, plant-based milk is likely to have a low PDCAAS or DIAAS due to the limiting amino acid contents of methionine, cysteine, and/or lysine. Hemp seed protein has lower PDCAAS values than soy milk but higher than almond and oat milk. hemp seed milk is higher fat milk than other plant-based milk.

Compared to cow's milk, plant-based milk has lower saturated fatty acids but a higher unsaturated fatty acid profile. It is reported that a food with a higher ratio of unsaturated fatty acids instead of saturated fatty acids is beneficial for LDL cholesterol.

Colour is a feature that should be considered in the formulation development of new plant-based milk. Especially if the product is to be recommended as a cow milk replacer, it should have similar values to cow's milk. The white index calculated using colour values (L, a, b) can be used as a good indicator for comparison.

2.5 HEALTH BENEFITS OF HEMP MILK

Hemp seeds offer medical benefits in four areas: cardiovascular, mental health, dermatological, and immune system. Furthermore, they have a low glycemic index for diabetics due to their low carbohydrate content. The most significant health benefits include enhanced energy, cholesterol management, improved circulation, immunological and neurological system strengthening, rapid recovery from injuries, reduced stress, and decreased symptoms of atopic dermatitis and allergies. Hemp's high soluble fiber content aids in the health and cleanliness of the digestive tract. They have anti-cancer qualities due to their high concentration of plant sterols and antioxidants and help prevent diseases such as Parkinson's and Alzheimer's and improve memory. Hemp is also advised for the treatment of premenstrual syndrome (menopause) because to its gamma-linoleic acid concentration, as well as for joint illness and stress reduction due to its magnesium and high-quality proteins. It is useful in maintaining nervous system homeostasis, as well as treating skin conditions such as psoriasis and eczema.

Hemp seed milk has the advantage of being free of milk proteins and lactose, making it ideal for persons who are allergic to or intolerant to cow's milk. The study found that consuming hemp seed milk decreases blood triglyceride and cholesterol levels. Based on what has been said, hemp and hemp seed milk are regarded to have promise as a nutritious and beneficial non-dairy alternative.

(Besir et al., 2022)

Table 1. Comprehensive Benefits of Hemp

HEMP FEATURE	BENEFITS
Protein	The high nutritional value of hemp protein can help one to reach the adequate intake of protein in a vegetarian diet without animal sources.
Essential fatty Acids	A vegetarian diet has limited sources of EFAs, especially omega 3. The advantageous ratio of n6/n3 in hemp oil can help to obtain a balanced intake of polyunsaturated fatty acids.
Calcium	There are limited sources of calcium in a vegetarian diet without milk and dairy usage. Using various plant foods including hemp can help to reach the RDA for calcium.
Iron	In a vegetarian diet, there are only low-bioavailable sources of iron so different iron-rich foods must be used, and hemp can have a high content of this mineral
Fiber	Even if all plant-based foods are rich in dietary fibers, the functional properties of insoluble fiber from hemp can stimulate its consumption.

Phytochemicals	The high content of bioactive molecules with health effects can stimulate the consumption of hemp in a plant-based context.
Environmental impact	Considering the wide motivation for a vegetarian diet adoption, the low impact of hemp cultivation can prompt the use of hemp as an eco-friendly plant-based source
Versatility	Hemp can have promising features, and the seeds can be employed in various industrial productions including some products of interest for a vegetarian diet such as plant-based milk and meat alternatives. Moreover, employed as a fortifier, hemp derivatives can be used in supplement and bakery products.

(Sabaté & Ratzin-Turner, 2001)

2.6 DEFECTS/DISADVANTAGES OF HEMP MILK

Hemp beverage possessed a hay-like odour and astringent taste. With GC-O analysis, the highest alcohols and ketones content were determined in hemp drinks. These compounds may have affected odour by causing hay-like characteristics, as was described by the sensory analysis. Hemp drink also stands out with p-cymene (spicy, phenolic, clove) content, known as phenolic and spicy in odour. This compound may increase astringency and intensify the aftertaste. Furthermore, it was suggested that the specific hemp flavour was caused by different terpenes in that study.

Phase separation is another defect commonly found in Hemp milk. Ultrasound and homogenization process applied to hemp seed milk prevent or delay the phase separation problem without adding additives. It was reported that the lowest amount of serum separation achieved was found applying homogenization (40 MPa) combined with ultrasonication (20% amplitude for 2 minutes). When it was examined the effects of ultrasound and homogenization process separately, the optimum parameters of the ultrasound process was determined as 50.2% amplitude and 6.40 minutes. Furthermore, it was reported that ultrasound application increased the L^* value of hemp seed milk and decreased the amount of syneresis and serum separation statistically. (Beşir et al., 2019)

Hemp protein is known for its low emulsifying capacity due to its compact structure and poor water solubility. There have been attempts to modify the native structure of hemp protein for improving the emulsifying capacity, for example, enzymatic hydrolysis. An alternative method to modify the structure of proteins is pH shift, namely, to expose native protein to an extreme pH condition that induces partial unfolding followed by incubation in a neutral pH environment to allow partial refolding. Such a process disrupts the tertiary structure and generates a molten globule conformation that has a markedly enhanced

amphiphilicity. Hence, pH shift has been successfully applied to the treatments of legume proteins, including soy protein and pea protein for improved emulsifying properties.

(Wang et al., 2018)

No study was found in which antinutritional factors were detected in direct hemp seed milk. However, antinutritional components such as phytic acid, trypsin inhibitors, condensed tannins, cyanogenic glycosides, and saponins have been reported in hemp seeds as in other edible seeds.

(Farinon et al., 2020)

2.7 EXTRACTION OF HEMP MILK

Hemp seeds are marketed as hulled or whole. Hemp seed milk is made by homogenizing ground seeds in water (1:5 w/v), and the filtered milk, is generally heated for extended shelf life. In consequence, there have been many attempts to improve the emulsifying capacity of hemp protein by altering its original structure, such as 1) enzymatic hydrolysis as they have noticed an improvement in the protein solubility of hemp protein after the application of limited enzymatic hydrolysis by trypsin enzyme, but there was a remarkable decrease in the emulsifying activity index; 2) acylation, it has been reported that acetylation treatment improved the emulsifying activity indexes of hemp protein and aided in the formation of more soluble protein aggregate. Further, hemp seed proteins have low heat stability and are affected and started to aggregate when exposed to heat treatment at 80°C or above for 10 min.

First, the production is carried out using seeds that have low levels of tetrahydrocannabinol (THC) (less than 0.3%). The hemp seeds have been mixed with water and hydrated into a slurry by absorbing water and being softened. After being waited in the mixing tank for a certain period, the hemp seed slurry is applied to wet milling, and the resulting slurry is recovered and circulated back to the mixing tank over and over for around ten minutes until reaching the desired consistency and maximum yields. Then a filtration step removes the shells, fibers, and other solid materials. After that, the filtered hemp slurry or milky mixture with about 16% solids is homogenized for increasing the stability and resulting white creamy hemp seed milk. To stop the vitamins and oil oxidation, the hemp seed milk temperature was reduced in a cooler to under 10°C. Water, emulsifiers, stabilizers, sweeteners, and/or flavourings can be added. Finally, to extend the hemp milk's shelf life, aseptic processing involves sterilizing the product using an ultra-high temperature (UHT) treatment followed by sterile packaging. The advantage of the current invention is that it

provides milk that is liquid, white, and high in essential fatty acids Omega 3 and 6, low in carbohydrate, and has a good flavour. (Hemp food product base and processes)

In another method to obtain hemp seed milk that has a stable colour (never becomes grey) which is free of bitter taste, and can be pasteurized and sterilized, the whole hemp seeds in a ratio of 15-25% are soaked for 5-12h in water at a temperature between 5-35°C. Then, water is used for the extraction, and the resulting raw hemp seed milk is separated from the remained solids of the hemp seeds . A lot more plainly, hemp seed milk has been produced in two different ways, 1) the hemp seeds are immersed in water for 5-12h and then grounded around 15 minutes. Afterward, milk extraction is carried out at a temperature below 80°C, or 2°C. Hemp seeds are crushed at temperatures ranging from 0°C to 80°C. After the extraction, obtained milk for 3-4 hours, additional cold extraction at a temperature between 0-10°C has been carried out. Latterly, the obtained extracted hemp seed milk is separated into peeling part which contains proteins and other hemp seeds particles, and raw hemp seed milk which can be sold directly to consumers or processed further, like applying homogenization, the addition of emulsifiers and Flavours, pasteurization (60-93°C), or sterilization at a temperature above 100°C. However, it is established that when performing mashing and extraction of milk at temperatures of 80°C and above, the coagulated proteins during the process do not extrude from the beginning; however, they are separated at the end of the line with other divided solid substances. (Method for producing hemp milk) (Farinon et al., 2020)

2.8 PREPARATION OF FROZEN DESSERT

2.8.1 STUDY 1

Ingredients used for preparing the best options of vegan ice cream production: roasted almond, almond syrup, saccharose, glucose, inulin, pea protein , emulsifier E471 , milk powder without lactose (Mlekovita) – eliminate from the final formula, LBG (Fooding Shanghai), xanthan gum.

Preparation : All ingredients were weighed separately with a determined formula pasteurization process of the ice cream mix at 95°C per 1.5 min - the mix was cooled 25°C, and frozen through 40 - samples placed into a plastic box and hardened at –18°C for 24 h - a control sample of ice cream without any stabilizers (WOS)

Key Findings

- pH similar to that of milk ice cream – 7.7 to 7.8 , with stabilizer and without – no significant difference, almond milk and pea protein – may have contributed to increase in pH
- Measured overrun values of prepared ice cream were among 26.56 and 32.02% for WS and WOS respectively – depends on density
- Low overrun compared to milk – milk proteins
- locust bean gum and xanthan gum as stabilizers to ice cream caused elongation in the melting time, also inulin – prebiotic and elongation in melting
- Sensory – no noticeable difference
- important impact on the prepared ice cream samples, was given by the almond drink and the pea protein. The flavour sensation was considered as acceptable and smooth. But, for the sample with locust bean gum and xanthan gum addition, this indicator was recognized as more aromatic

(Study of the properties of vegan ice cream based on almond drink)

Table 1. Final formula of vegan ice cream

Tabela 1. Finalna receptura lodów wegańskich

Ingredients Składniki	Weight Udział wagowy [g]	Percentage Udział procentowy [%]
Almond drink Napój migdałowy	587.36	73.42
Almond syrup Syrop migdałowy	64.00	8.00
Inulin Inulina	96.00	12.00
Pea protein Białko grochu	48.00	6.00
Emulsifier E471 Emulgator E471	3.20	0.40
Locust bean gum (LBG) Mączka chleba świętojańskiego (LBG)	0.80	0.10
Xanthan gum Guma ksantanowa	0.64	0.08

2.8.2 STUDY 2

Ingredients: Almond seeds, agave syrup, and vanilla, High quality peeled hemp, psyllium and pectin fibre

Almond Milk Preparation - The almond seeds (300 g) soaked overnight at 4 °C in 900 mL of distilled water. For milk extraction, the almonds were mixed with distilled water (1:10 g/v ratio), ground in a blender and filtered

Hemp Milk Preparation -. The dehulled seeds of hemp (300 g) were soaked for 10 h at 20 °C in 900 mL of distilled water. For milk extraction, the hemp seeds were mixed with distilled water (1:10 g/v ratio), ground

The milk heated to 80 °C, then homogenized and frozen in a 1.5-L ice cream maker, the samples were frozen and stored in a refrigerator at −22 °C for subsequent analysis

Key Findings

- Hemp milk with added pectin has higher titratable acidity values as the percentage of added pectin increases
- Hemp milk samples had a fat content relative to milk fat, which was around 3.5%- a lower increase in hardness
- The control sample with hemp milk had a content of 2.04% and the sample with 8% pectin fibres had a higher protein content, 2.79%.

(Leahu et al., 2022)

2.9 Determination of the nutritional composition of vegetable milks

2.9.1 Protein content:

The Kjeldhal technique was used to determine the total protein content of each milk sample. The first step is to demineralize the organic matter using sulfuric acid, followed by a second step of alkalizing the reaction products. The coefficient of conversion of organic azote to proteins is 6.25.

2.9.2 Lipid content:

It was determined by dissolving milk in a solvent, which in this case is ether diethylique, and then evaporating the solvent.

2.9.3 Sugar content:

The sugar content was determined using spectrophotometry.

2.9.4 Fiber content:

The fibre content was determined using AOAC 2015 standards. A portion of 2 g lyophilized milk is placed in 200ml of 1,25 H₂SO₄ and boiled for 30 minutes. The solution was then placed in a funnel bucher with a muslin fabric. The residue was filtered and then washed with hot water to remove the acid. The residue was then placed in 200 ml of 1.25 % NaOH and boiled for 30 minutes before being filtered. The residue is then placed in a humidity extraction oven until it reaches a constant weight.

2.9.5 Calcium content:

The calcium content was determined using spectrophotometry.

2.9.6 Determination of the antioxidant activity and the total polyphenol content Extraction:

For the extraction step, a total of 25 mL of an 80 % methanol solution was added to 5g of each lyophilized sample, followed by agitation for 24 hours at room temperature. The tubes were then centrifuged for 5 minutes at 6000 tr / min. The supernatant (extract) was collected after centrifugation for analysis.

2.10 Physico-chemical properties of vegetal milk

pH: The pH of the four milks was determined by diluting them 1:10 in pure water and measuring them using a pH meter. For each studied sample, all measurements taken at ambient temperature.

Colour: Measured with a colourimeter, and the results were expressed in terms of the chromatic space L * a * b * as defined by the International Commission on Lighting (CIE) in 1976. The three coordinates represent the colour's clarity (L *), its location between red

and green (a *), and its position between yellow and blue (b *). The whiteness index was calculated using the following formula:

$$WI = 100 - \sqrt{((100 - L)^2 + a^2 + b^2)}$$

(Jemaa et al., 2021)

Oxidation stability: Freshly prepared hemp milk was placed in Teflon-capped test tubes and incubated in the dark at 37 °C (to accelerate the reaction) for up to 8 days. To prevent microbial growth so that only chemical oxidation was assessed, sodium azide (0.02%, w/v) was added to all samples. Oxidative stability was evaluated by using both peroxide value (PV) and thiobarbituric acid-reactive substances (TBARS)

(Wang et al., 2018)

2.11 Determination of physicochemical properties

Viscosity : The apparent viscosity of the mixtures was evaluated after 24 h of aging using a Brookfield viscometer. Samples were tested in triplicate at 4° at a speed of 20 rpm.

Overrun : The formula for overrun is as follows:

Overrun % = [(weight of unit volume of mix) - (weight of unit volume of ice cream)]/(weight of unit volume of ice cream) × 100.

Meltdown rate: The meltdown rate was determined by carefully cutting the plastic cups from the ice cream samples (pre-weighed as 70 g), placing the ice cream onto 1-mm stainless-steel mesh over a funnel and measuring the amount of ice cream that drained into the graduated cylinder at 20 ± 0.5°C. Meltdown was expressed as the volume melted over the time required to collect 20 mL of melt. The measurements were performed in triplicate.

Texture analysis: Ice cream samples stored at -20°C were tempered to -10°C for 24 h before analysis. The ice cream, which remained in the plastic cup, was penetrated in two places on its largest smooth surfaces, and four measurements were recorded.

(Ismail et al., 2013)

3. INGREDIENTS AND EQUIPMENTS

3.1 INGREDIENTS

Hemp seeds and Hemp hearts

Hemp seeds are the seeds of the hemp (*Cannabis sativa*) plant. Technically, the industrial hemp plant is the same as the cannabis (marijuana) plant. Hemp hearts are the inner part of hemp seeds. They are softer than hemp seeds and have almost the same nutritional value except for lacking Fiber. Here both hemp seeds and Hemp hearts were used for the formulation of ice cream and the best option was chosen. Hemp seeds were procured from a local manufacturer while hemp hearts were bought online and were of the brand 'emptyful hemp hearts'.

Coconut cream

Unsweetened Coconut cream was used as the major fat source in hemp milk ice cream and has almost 24% fat content. Coconut cream was purchased online from the brand Urban platter. Coconut cream helped in providing the creamy texture characteristic of ice cream as well as helped to mask the beans Odour of hemp.

Sweetener

Sugar was used as the main source of sweetness. Sugar was procured from margin free supermarket, Mannuthy, Thrissur. Complete replacement of sugar did not provide the required sensory profile and thus a part of it was replaced by glucose syrup. Glucose syrup helped in preventing crystallization and also to provide a smooth texture to the Product. Liquid glucose was procured from Best Tradings, Thrissur and was of the brand 'Ck's Liquid Glucose'.

Pectin

Pectin helped to create a smooth, creamy texture in vegan ice cream, similar to traditional ice cream. It thickens the mixture and provides a gel-like structure. It also stabilizes the ice cream, preventing ice crystals from forming and giving it a more even consistency. Ck's pectin powder was used for this purpose.

Glycerol monostearate

GMS helps to emulsify the mixture, combining oil and water-based ingredients together, creating a smooth and creamy texture. It helped to modify the texture of the ice cream, creating a more scoopable and creamy consistency. GMS was bought online and was of the brand 'Bake King'.

Water

Water forms a major portion of ice cream and is used for extraction of hemp milk from hemp seeds and Hemp hearts. Distilled water was used for the same.

3.2 EQUIPMENTS

Ice cream maker

Agaro Maple ice cream, sorbet, slush and frozen yoghurt maker of 1.2 litres capacity was used for lab scale formulation and standardization of hemp milk ice cream. The equipment consists of a double walled mixing bowl. The inner walls are filled with refrigerant which must be frozen prior to use. A motor run paddle gives the churning action, scraping and mixing the contents in the bowl. 10-30 minutes of mixing is done depending on the product and the raw materials used.



Fig.1

4. FORMULATION OF SAMPLES

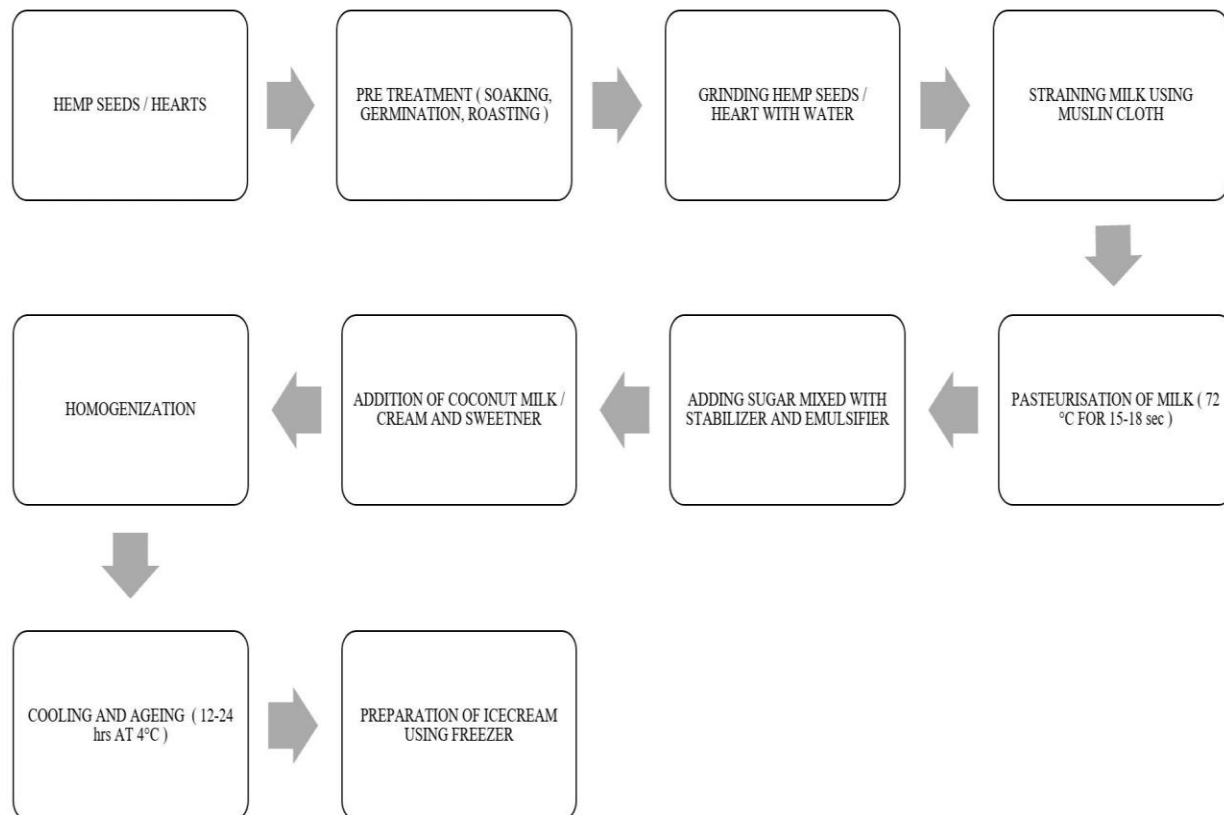


Fig.2

4.1 Pre-treatment methods

Pre-treatment has the benefits of enhancing sensory attributes like taste and appearance of hemp seeds and hemp hearts. Also, these methods have the possibility of lowering down the amount of residual THC in the final products. Three types of pre-treatment procedures, namely, soaking, germination, and roasting, were investigated in terms of the efficiency for hemp seeds and hearts processing.

4.1.1 Soaking (For hemp seeds)

Soaking entails placing the seeds in lukewarm water for about 10-12 hours. This method partially opens the hull but does not fully remove it. The final product had a greyish hue that proved visually unappealing. Furthermore, the treatment retained an undesirable hay-like or beany flavour associated with hemp seeds and resulted in an undesirable product.

4.1.2 Germination (For hemp seeds)

Germination was done by first placing the seeds in a tray with lukewarm water for 6 hours and then transferring them in a damp cloth for sprouting at night. This technique provided only partial debarking of the hull and a small increase in the seed colour. In spite, the germinated seeds still possessed a very evident hay-like or beany flavour with a bitter note at the back of the throat, and therefore, this method cannot be recommended for increasing palatability.

4.1.3 Roasting (For hemp hearts)

Heating was done on hemp hearts which are essentially the seeds but without the hulls, by heating the seeds at a medium flame for 7 minutes. As applied it substantially enhanced the sensory characteristics of the product. The roasting process helped to remove the bitter hay like or beany flavour to an extent as well as contributing to a nice nutty flavour and smell. Therefore, roasting was observed to be the best pre-treatment process for enhancing the organoleptic acceptability of hemp hearts.

4.2 Extraction of Hemp Milk

Hemp milk seems to be a very promising ingredient which can be included in numerous food products especially for vegan and lactose intolerance consumers. The hemp hearts after being soaked are then blended with a particular proportion of water. This blending normally occurs at high blend rates for the purpose of giving the seeds a thorough mixing in the blender so that the oils, proteins and other soluble nutrients from the seeds can dissolve into the liquid media. The blending time and water to hemp ratio are changed with a view of achieving the required viscosity and nourishes the hemp milk.

Hemp milk was produced by blending the hemp seeds with a small amount of warm water. A 1:10 ratio of hemp to water was found to be optimal for obtaining a product with superior sensory characteristics. This ratio provided the aqueous component of the emulsion with the desired smooth mouthfeel and optimised the taste and potentially minimised unpleasant 'coating' afterwards.

The blended mixture is then filtered using a fine mesh, muslin cloth or a cycle filtration system to isolate the solid leftovers. This step is to make sure that the final product, the hemp milk is smooth and creamy without any coarse particles. The solid residue was separated from the mixture using a muslin cloth to give a smooth and palatable hemp milk product.



Fig.3



Fig.4

4.3 Pasteurization of Hemp Milk

For reasons of achieving microbiological safety and extending the shelf life of the final product, the extracted hemp milk was pasteurized. They included heating the milk to 72°C for 15–18 seconds and then cooling the milk rapidly to the room temperature. This mode of getting heat was effective in eliminating the negative effects contributed by microorganisms without altering the nutritive and organoleptic values of the milk. Of particular concern in pasteurization process, one feature was observed that high temperatures distorted and excluded the right consolidated state of milk and cause agglomeration of milk.

4.4 Preparation of ice cream mix

This is tasks involve preparing the ice cream mix from particular ingredients in proportion to have the right texture, flavour and stability. There are five key ingredients in the ice cream mix; water or hemp milk, sweeteners, natural sugars, stabilizers and emulsifying agents. Respective ingredients are weighed and measured in accordance with the specification of the product to be developed. The basic component typically a hemp milk or other liquid is introduced into the mixing vessel. This provides the medium for conveying all the other ingredients and helps in the creation of the inadequate texture of the finished product.

The sweeteners are dissolved into the liquid base in order to get good distribution of it in the mixture. Stabilisers is added for the texture of the compound as well as for enabling prevention of formation of ice crystals during freezing. Emulsifiers used to enhance the stability of the mix and to produce a cream like texture through preventing separation of fat and water.

Pre-heating and homogenization are types of heat treatment that is used to change the valued properties of a material. The mixture is usually warmed up to about 40- 45 degrees Celsius to allow ease of mixing. In homogenization a pressure is applied on the mix to control the size of fat globules and to avoid separation. Thereafter, the mix is cooled to about 4°C as quickly as possible to avoid the growth of microbial in the milk. This is followed by binding at 4-12 hours in a cold condition. This aging also assists the stabilizers and emulsifiers undergoing a process of hydration whereby the viscosity and the whipping ability of the mix increases.

This ice cream mix is now ready to be considered for freezing and other processes subsequent to it, to make the final frozen dessert. It saves time and assures the product attains the desired sensory and structural characteristics that would make the consumers feel satisfied.

4.5 Preparation of ice-cream

The preparation of ice cream involves the use of a freezing bowl that is refrigerated to maintain the right temperature for the operation. The recommended time for refrigerating the bowl for at least 12–24 hours before the use of it. Pour pre-prepared ice cream mix into a refrigerated freezing bowl. Chilling the mix before freezing the results should ideally speed the process and result in the best possible product. Don't overfill the bowl, leave sufficient space so the mix can expand as air is mixed in during the churning.

The churning process should begin when the paddle inside the freezing bowl becomes round out of the machine. Securely attaching the paddle to the machine assembles the ice cream maker. So, the paddle churns the mixture to get a smooth and creamy mixture with air added in as well as freezing the mix evenly. During mixing make sure the paddle is properly aligned and locked in place so as to avoid operational issues.

Let the machine churn for 30 minutes or so. The freeze bowl will keep cold and the ice cream paddle will mix the cold ice cream continuously until the end of this time. This step is important to obtain the texture and consistency in the final product. After 30 minutes the ice cream should get to a semi solid state and can be consumed or frozen for a firmer texture later.

4.6 Experimental Design

Eight formulations of ice cream mix were developed, varying the amounts of sweeteners, stabilizers, and coconut milk/cream.

In Formulations A to D, 100 grams of tablets are sweetened with Sugar. The dynamics of Formulations E to F involve 100 grams of sugar and 15 ml of liquid glucose with the formulations exhibiting extra viscosity and sweetness. Whereas, the two formulated products, G and H, have sugar level slightly lowered at 80g and liquid glucose has been raised at 20 ml to improve its smoothness and make it less crystalline.

The Formulation A contains a stabilizer mix (1.25g) which gives the substance a stable structure and avoids formation of two separate phases. Stabilizer mix contains Guar gum, Mono and diglycerides of fatty acids, Sorbitan esters of fatty acids and Carrageenan. The second formulated B replaces the synthetic stabilizer found in formulation A with natural pectin in the proportion of 20 grams.

Both Formulations C and D contain 1.25 ppm of Glycerol Monostearate, an emulsifier with Formulation D containing an additional 1.25 ppm of pectin for stabilisation. Formulation E minimizes GMS at 0.75 ppm whilst retaining 1.25 ppm of pectin for a balanced emulsifying system. The pectin is formulated at 2.5 percent in formulations F and G and at 0.75 percent in formulation H; for additional stabilization and mouthfeel, GMS is incorporated (1.25 percent in formulation H only).

In Formulation A, 100 ml of coconut milk is used while in Formulation B, 100 ml of coconut milk is also use but both coconut . In Formulations C and D, 50 ml of coconut cream with a 20% fat content have been used to make it thicker.

In formulations E to H, the quantity of coconut cream is raised to 100 ml with in E and F as well as 125 ml in G and H, so as to allow for a higher fat content that imparts a creamier mouthfeel. Fat content also increased from 20% to 23%.

Each formulation involved blending the specified ingredients to form a homogeneous ice cream mix. The blends were homogenized and cooled to room temperature, and churned using an ice cream maker. The final ice cream samples were frozen at -18°C for 24 hours before analysis.

Table 2. Formulations for Ice Cream Mix Preparation

FORMULATION	SWEETNER	STABILIZER / EMULSIFIER	COCONUT MILK / CREAM
A	Sugar (100 g)	Stabilizer mix (1.25 g)	Coconut milk (100 ml)
B	Sugar (100 g)	Pectin (20 g)	Coconut milk (100 ml)
C	Sugar (100 g)	GMS (1.25 g)	Coconut cream (50 ml, 20% fat)
D	Sugar (100 g)	GMS (1.25 g) and Pectin (1.25 g)	Coconut cream (50 ml, 20% fat)

E	Sugar (100 g) and Liquid glucose (15 ml)	GMS (0.75 g) and Pectin (1.25 g)	Coconut cream (100 ml, 20% fat)
F	Sugar (100 g) and Liquid glucose (15 ml)	Pectin (2.5 g)	Coconut cream (100 ml, 20% fat)
G	Sugar (80 g) and Liquid glucose (20 ml)	Pectin (2.5 g)	Coconut cream (125 ml, 23% fat)
H	Sugar (80 g) and Liquid glucose (20 ml)	Pectin (0.75 g) and GMS (1.25 g)	Coconut cream (125 ml, 23% fat)

5. RESULT AND DISCUSSION

5.1 RESULTS OF SAMPLES

The sensory characteristics were assessed through a combination of Odour, flavour, and texture evaluations. The following observations were made:

5.1.1 Sensory Evaluation of Sample A

1. **Odour:** A persistent "hay-like" Odour was detected during the evaluation. Throughout the sample's analysis period, this feature was evident and persisted.
2. **Flavour:** The flavour profile of Sample A was predominantly marked by a **rancid** taste. This off-flavour was identified as unpleasant and could negatively influence consumer acceptability.
3. **Texture:** Participants reported an unfavourable mouthfeel in Sample A, which diminished the entire sensory experience and was considered unattractive.
4. **Taste:** A distinct **beany** taste was also noted, which indicate the presence of certain compounds associated with legumes or potential fermentation by-products.

These sensory attributes suggest that Sample A may have developed undesirable characteristics over time or due to certain processing conditions.

5.1.2 Sensory Evaluation of Sample B

1. **Odour:** Similar to Sample A, Sample B exhibited a persistent "**hay-like**" **Odour**. This Odour was detectable throughout the assessment period and may indicate an issue related to storage or processing conditions.
2. **Flavour:** The sample displayed a **rancid flavour**, which was unpleasant and indicative of potential spoilage or undesirable chemical changes within the sample.

3. **Texture:** In addition to the **unappealing texture** observed, Sample B demonstrated a **fluffy texture** that was attributed to an excess of **pectin** (approximately 8%). The changed texture, which was deemed undesirable, could have been caused in part by this increased pectin concentration.
4. **Taste:** Like in Sample A, a beany taste was also observed, possibly due to the inherent flavour of the raw materials or degradation during processing.

The combination of these sensory attributes suggests that Sample B suffers from similar quality issues as Sample A, with the added complication of altered texture due to excess pectin. To enhance the formulation and increase sensory attributes, further investigation of pectin content and its impact on texture is advised.

5.1.3 Sensory Evaluation of Sample C

1. **Aftertaste:** A significant reduction in **aftertaste** was observed in Sample C. This improvement indicates that certain unwanted residual tastes were reduced, resulting in a more agreeable sensory profile.
2. **Flavour:** The overall **flavour** of Sample C was notably improved. Sensory panellists observed a more pleasant and balanced flavour, indicating a favourable change in the formulation or processing method utilized.
3. **Phase Separation:** However, **phase separation** was observed during the analysis, which may indicate instability in the formulation. This could suggest issues related to emulsification or dispersion of ingredients.
4. **Beany Taste:** There was a **reduced beany taste** compared to previous samples, suggesting a reduction in the characteristic off-flavour associated with legumes or fermentation by-products.

5. **Appearance:** The appearance of Sample C was marked by a **grainy texture**, which could indicate a partial breakdown of the matrix or incomplete homogenization of ingredients during processing.

Overall, Sample C showed several positive changes, particularly in flavour and aftertaste, although the presence of phase separation and a grainy texture warrants further investigation to understand the underlying causes and improve the stability and consistency of the product.

5.1.4 Sensory Evaluation of Sample D

The sensory evaluation of **Sample D** indicated further improvements compared to previous samples, along with some persistent texture-related issues. The main findings are summarized below:

1. **Aftertaste:** A **reduced aftertaste** was observed in Sample D. This improvement indicates a more pleasant and less lingering sensory experience, which adds to total consumer approval.
2. **Flavour:** Sample D exhibited an **improved flavour** profile. Panelists noted a more balanced and refined taste, suggesting enhancements in formulation or processing techniques.
3. **Phase Separation:** There was a **reduction in phase separation** in Sample D, indicating improved formulation stability and uniformity compared to previous samples. This could be a result of better emulsification or ingredient integration.
4. **Beany Taste:** The **beany taste** was again reduced in Sample D, showing progress in reducing unwanted flavours from beans or fermentation byproducts.
5. **Appearance:** Despite improvements in flavour and aftertaste, Sample D still exhibited a **grainy appearance**. This persistent texture issue suggests that while other aspects of the product have improved, the textural consistency remains a

challenge, potentially due to incomplete ingredient dispersion or processing limitations.

In summary, **Sample D** showed significant improvements in flavour, aftertaste, and phase separation compared to earlier samples. However, the appearance is grainy.

5.1.5 Sensory Evaluation of Sample E

The sensory assessment of **Sample E** revealed several notable improvements in flavour and texture, along with one potential concern. The key observations are as follows:

1. **Sweetness:** **Sample E** was perceived as **overly sweet**, which may be due to an imbalance in the sweetness level relative to other flavour components. This excessive sweetness may have an impact on overall acceptance, especially among consumers looking for a more balanced product.
2. **Aftertaste:** One of the most significant improvements observed was the **significant reduction in aftertaste**. This suggests that the formulation may have been adjusted to remove residual, unpleasant flavours, so improving the entire sensory experience.
3. **Flavour:** The overall **flavour** of Sample E was found to be **improved**, with a more harmonious and pleasant taste profile. This suggests that formulation adjustments or processing changes have contributed to a more balanced and appealing flavour.
4. **Texture:** **Sample E** demonstrated a **smooth texture**, which was well-received by sensory panelists. This improvement in texture enhances the overall mouthfeel and could contribute positively to the product's appeal.
5. **Beany Taste:** The **beany taste** was again **reduced** in Sample E, indicating continued success in eliminating this off-flavour, which is commonly associated with leguminous ingredients.

In conclusion, **Sample E** exhibited a significant reduction in aftertaste and beany flavour, along with improvements in texture and overall flavour. However, the overly sweet taste could be a drawback.

5.1.6 Sensory Evaluation of Sample F

The sensory evaluation of **Sample F** revealed several positive improvements, along with some areas that may require attention. The key findings are outlined below:

1. **Sweetness:** **Sample F** was perceived as **overly sweet**, similar to previous samples, which could impact consumer preference. To get a more balanced flavour profile, this characteristic may need to be adjusted.
2. **Aftertaste:** The **aftertaste** in Sample F was **negligible**, representing a notable improvement over other samples, where lingering flavours had previously been a concern. This suggests optimization in formulation or processing to minimize residual aftertaste.
3. **Flavour:** The **flavour** of Sample F was deemed **improved**, with a more pleasing and balanced taste compared to earlier samples, indicating that the formulation changes were successful.
4. **Texture:** The sensory panelists expressed a favorable opinion of Sample F's smooth and creamy texture. The overall mouthfeel and customer experience are improved by this notable improvement in textural quality.
5. **Stability:** The sample exhibited a **stable scoop**, suggesting that the product maintained its form and consistency, which is an important indicator of physical stability and handling.
6. **Beany Taste:** The **beany taste** was significantly **reduced**, further indicating the success in eliminating off-flavours associated with legumes or fermentation by-products.

7. **Appearance:** Compared to earlier samples, Sample F had a less noticeable grey tint, which could increase the product's market attractiveness.

In conclusion, **Sample F** demonstrated significant improvements in flavour, texture, aftertaste, and appearance. However, the overly sweet taste remains a concern and should be addressed to achieve a more balanced and acceptable product.

5.1.7 Sensory Evaluation of Sample G

The sensory evaluation of **Sample G** demonstrated several key improvements, particularly in flavour, texture, and overall product stability. The main findings are as follows:

1. **Aftertaste:** **Sample G** exhibited a **negligible aftertaste**, a significant improvement over previous samples, suggesting that the formulation successfully minimized lingering undesirable flavours.
2. **Flavour:** The **flavour** of Sample G was notably **improved**, with a more refined and balanced taste profile. This enhancement suggests that the formulation or processing changes were successful in optimizing the flavour.
3. **Texture:** Sample G displayed a **smooth texture**, which was well-received during sensory testing. This contributes positively to the overall mouthfeel and consumer experience, making it more appealing to the target audience.
4. **Stability:** The sample also demonstrated a **stable scoop**, and retained its shape and structure throughout handling, ensuring product stability and ease of use.
5. **Appearance:** The product's appearance improved with less grey tint compared to previous samples. This enhancement implies more control over colour constancy, which is critical for consumer approval.
6. **Palatability:** **Sample G** was noted to have **improved palatability**, meaning that it was more enjoyable and acceptable overall. This is a significant aspect in determining consumer satisfaction and marketability.

7. **Melting Resistance:** One of the significant improvements observed was **increased melting resistance**, which suggests that Sample G has better structural integrity under varying temperature conditions, enhancing its shelf-life and handling characteristics.

In conclusion, **Sample G** demonstrated notable improvements across multiple sensory attributes, including flavour, texture, aftertaste, and stability. The increased melting resistance and improved palatability further indicate that Sample G has the potential for higher consumer acceptance. However, continued optimization may be necessary to fine-tune aspects like sweetness or specific textural features, depending on the target market.

5.1.8 Sensory Evaluation of Sample H

The sensory evaluation of **Sample H** demonstrated several notable improvements in both flavour and texture, as well as enhanced product stability and appearance. The key findings are as follows:

1. **Aftertaste:** **Sample H** exhibited a **negligible aftertaste**, marking a significant improvement over earlier samples where lingering flavours were more pronounced. This reduction in aftertaste shows that the formulation or processing was optimized to improve the overall sensory experience.
2. **Flavour:** The **flavour** of Sample H was **improved**, with a more balanced and pleasing taste profile, indicating effective adjustments in the product's composition.
3. **Texture:** Sample H had a **smooth and creamy texture**, contributing to a desirable mouthfeel. The smoothness was well-received during sensory testing, enhancing the overall consumer experience and palatability.
4. **Stability:** The **stable scoop** observed in Sample H indicates that the product retained its shape and consistency, a critical factor for both storage and ease of use.

5. **Appearance:** A **significant reduction in the grey tinge** was noted in Sample H, which substantially improved the visual appeal of the product. This suggests better control over colour and formulation, contributing to enhanced consumer acceptance.
6. **Palatability:** The **improved palatability** of Sample H reflects the combination of its improved flavour, smooth texture, and reduced aftertaste. Overall, the product was found to be more enjoyable and acceptable to the sensory panelists.

In summary, **Sample H** demonstrated significant improvements in key sensory attributes, including flavour, texture, aftertaste, appearance, and palatability. These changes suggest that the product formulation has been optimized for greater consumer appeal. Continued attention to potential adjustments in sweetness or other flavour components may further enhance the product's overall acceptability.

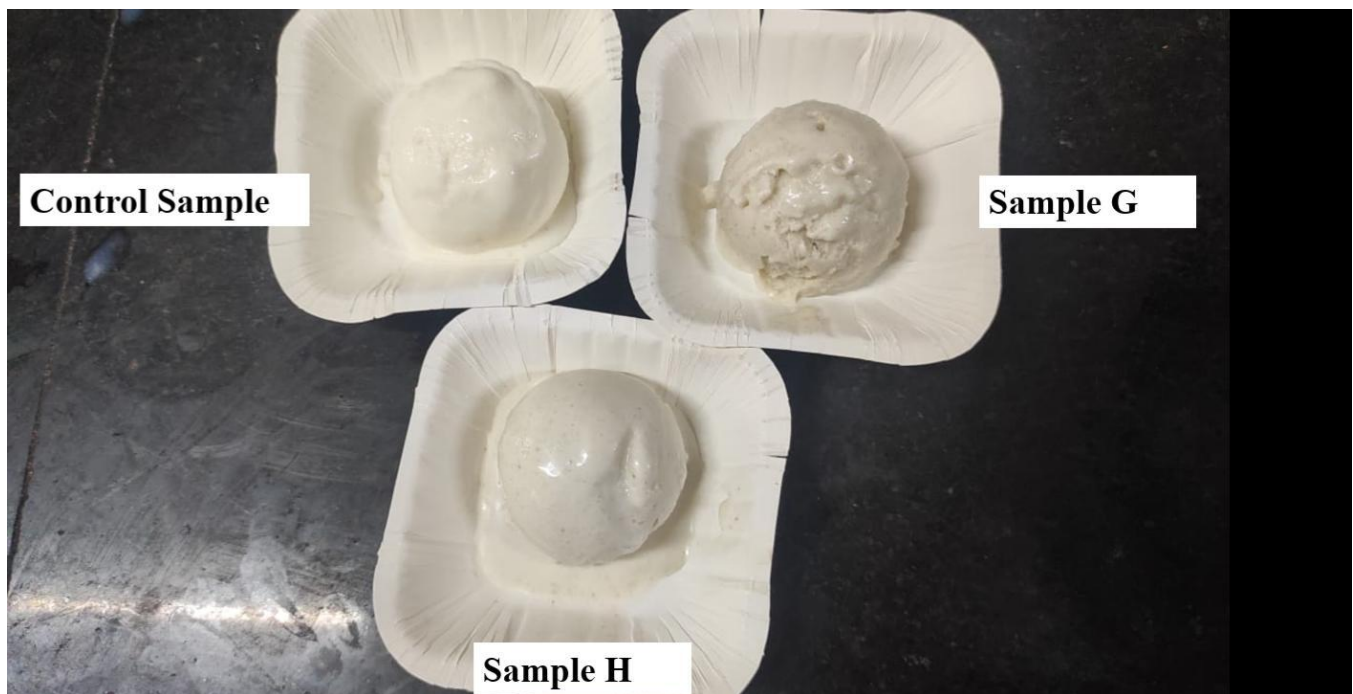


Fig.5

Best formulation for the preparation of hemp milk ice cream

Sugar :80g

Liquid glucose :20ml

Pectin - 0.75g

GMS - 1.25g

Coconut cream :125 (23% fat)

5.2 INGREDIENT BREAKDOWN

Effect of liquid glucose

The addition of liquid glucose has the potential to complement the inherent sweetness profile of the hemp milk, enabling formulators to achieve the desired level of sweetness without over-relying on other sweetening agents. Additionally, the freezing point depression and viscosity-enhancing characteristics of liquid glucose could prove advantageous in creating a smoother, creamier texture, which is particularly relevant for hemp milk-based ice creams. Furthermore, the stabilizing effect of glucose on emulsions could help prevent undesirable separation or graininess in the hemp milk-based ice cream mix, resulting in a more stable and consistent final product.

Effect of stabilizer and emulsifier

The incorporation of GMS and pectin in the hemp milk ice cream formulation plays a critical role in stabilizing the emulsion and enhancing the textural properties of the final product. GMS functions as an emulsifier, helping to stabilize the interface between the fat and water phases in the ice cream mix, preventing unwanted coalescence and separation, and contributing to a smooth, creamy mouthfeel. Pectin acts as a thickening and gelling agent, improving the body and texture of the ice cream by increasing the viscosity of the continuous phase and contributing to a more stable, uniform structure. The synergistic

effects of GMS and pectin work to mitigate issues like iciness, graininess, and destabilization that can arise in plant-based ice cream formulations.

Effect of coconut milk and coconut cream

The addition of coconut cream to the hemp milk ice cream formulation significantly enhances its textural and rheological properties. Coconut cream, with its high fat content, contributes to a richer and more indulgent mouthfeel. The fat globules in coconut cream act as emulsifiers, stabilizing the ice cream mixture and preventing the formation of large ice crystals. This results in a smoother, creamier texture with a reduced icy mouthfeel. Furthermore, coconut cream imparts a subtle coconut flavour, adding a unique dimension to the overall sensory experience. The combination of hemp milk and coconut cream creates a harmonious blend of flavours and textures, resulting in a truly delightful vegan frozen dessert.

Table 3: Sensory Profile of Prepared Samples

SAMPLE	APPEARANCE	TEXTURE	TASTE	SMELL	REMARKS
A	Exhibited an undesirable greyish cast and appeared rough	Unfavourable	Beany, Rancid flavour	Hay-like	• Persistent "hay-like" odour • Phase separation observed
B	Showed a dull grey undertone and appeared rough	Unfavourable, Fluffy	Beany, Rancid flavour	Hay-like	• Persistent "hay-like" odour • Phase separation observed

SAMPLE	APPEARANCE	TEXTURE	TASTE	SMELL	REMARKS
C	Grainy and reduced grey tint	Grainy	Reduced beany flavour	Reduced Hay-like odour	- Improved flavour - Phase separation observed
D	Grainy and reduced grey tint	Grainy	Reduced beany flavour	Reduced Hay-like odour	- Improved flavour - Reduced phase separation
E	Less grey tint	Smooth	Overly sweet	Reduced hay-like odour	- Phase separation was significantly diminished - Possesses a low melt resistance
F	Less grey tint	Smooth & creamy	Improved flavour	Reduced hay-like odour	- Improved flavour - Stable scoop - Possesses a low melt resistance
G	Reduced grey tint	Smooth	Improved flavour	Odour was acceptable, but not ideal	- Improved flavour - Stable scoop - Increased melting resistance
H	Colour clarity improved	Smooth & Creamy	Improved and more acceptable flavour	Odour was acceptable, but not ideal	- Stable scoop - Reduced grey tint - Possesses superior melt stability

5.3 HEDONIC SCALING ANALYSIS

Sample G and Sample H were selected for a sensory evaluation involving a panel of 20 evaluation panelists. The aim of the evaluation was to determine the consumer acceptance and preference for these two samples based on their sensory attributes. The sensory evaluation was conducted using a 9-point hedonic scale, where the panelists were asked to rate the samples on the following attributes: Appearance/Colour, Taste, Smell/Odour, Texture/Mouthfeel, and Overall Acceptability. The participants were instructed to indicate their level of liking or disliking for each attribute, with 1 representing "Dislike extremely" and 9 representing "Like extremely."

Result

Appearance/Colour

The Appearance/Colour of Sample H was rated significantly higher than Sample G, with scores of 178 and 124, respectively. This represents a 43.5% difference in favor of Sample H, indicating a stronger preference for the visual appeal of this sample.

Taste

The Taste attribute showed a similar trend, with Sample H scoring 172 compared to 123 for Sample G, a 39.8% difference. This suggests that the taste of Sample H was more preferred by the panelists.

Smell/Odour

The Smell/Odour of Sample H was also rated higher, with a score of 159 compared to 120 for Sample G, a 32.5% difference. This indicates that the aroma and odour of Sample H were more appealing to the assessors.

Texture/Mouthfeel

The Texture/Mouthfeel of Sample H was the most significant difference, with a score of 176 compared to 106 for Sample G, a 66.0% higher rating. This implies that the texture and mouthfeel of Sample H were substantially more desirable.

Overall Acceptability

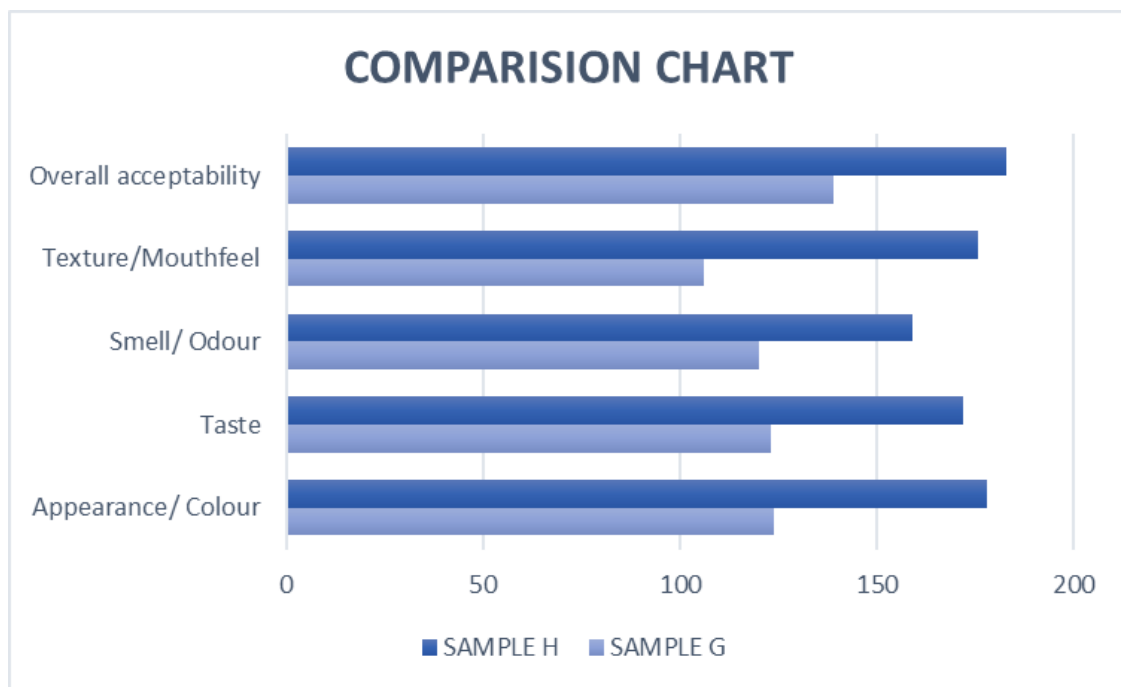
The Overall Acceptability score for Sample H was 183, compared to 139 for Sample G, a 31.7% difference. This suggests that the overall acceptance and preference for Sample H were significantly higher than for Sample G.

Conclusion

The hedonic scaling analysis revealed that Sample H was consistently rated higher than Sample G across all sensory attributes, with the most pronounced difference in Texture/Mouthfeel. The overall acceptability of Sample H was also significantly higher, indicating a stronger preference among the panelists.

Table 4: Hedonic Scaling Result of Sample H and G

CATEGORY	SAMPLE H	SAMPLE G
Overall Acceptability	180	135
Texture/ Mouthfeel	175	100
Smell/ Odour	150	120
Taste	170	120
Appearance/ Colour	180	120



6) CONCLUSION

The preparation of frozen desserts using hemp milk signifies a meaningful step forward in the development of sustainable, nutritionally balanced, and inclusive food options. This research has demonstrated that hemp milk, derived from the nutrient-rich seeds of the *Cannabis sativa* plant, possesses the functional properties required to produce high-quality frozen desserts. These include its ability to serve as a stable base for emulsification, its favorable freezing characteristics, and its capacity to contribute to a desirable texture and mouthfeel. The sensory evaluation results suggest that frozen desserts made from hemp milk can meet or exceed consumer expectations for taste, texture, and overall acceptability.

Hemp milk offers a unique combination of health benefits, including being a source of essential omega fatty acids, high-quality plant proteins, and vital minerals. Its naturally lactose-free, allergen-friendly, and vegan attributes make it a suitable choice for a broad spectrum of consumers, including those with dietary restrictions or ethical considerations. Moreover, its production has a lower environmental impact compared to conventional dairy milk, aligning with the increasing demand for eco-conscious food systems.

This study has not only highlighted the feasibility of creating frozen desserts from hemp milk but also opened avenues for further exploration in the broader field of plant-based food innovation. The development of these desserts has implications beyond the immediate product; it contributes to the larger movement toward diversifying food sources, enhancing sustainability, and reducing the environmental footprint of the food industry. The research also establishes a foundation for future studies focused on optimizing flavour profiles, improving shelf-life stability, and scaling up production for commercial applications.

Furthermore, this work underscores the importance of addressing consumer preferences and market trends when developing novel food products. As awareness grows around the

health and environmental benefits of plant-based diets, hemp milk-based frozen desserts can serve as a bridge between traditional dietary habits and modern food innovation. Their ability to satisfy indulgent cravings while adhering to principles of nutrition and sustainability makes them a promising addition to the market.

In conclusion, the preparation of frozen desserts from hemp milk represents a harmonious blend of science, innovation, and sustainability. This study has demonstrated the immense potential of hemp milk to serve as a cornerstone in the future of plant-based foods. By integrating this versatile ingredient into popular products like frozen desserts, we take a step closer to meeting the dual goals of consumer satisfaction and environmental responsibility. This work lays a robust groundwork for continued research and development, fostering a food system that is both nutritionally enriching and ecologically sustainable.

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